6.6 Single shot multibox detection

- in Section 6.3–Section 6.5, we introduced bounding boxes, anchor boxes, and multiscale object detection
- now, we are ready to use such background knowledge to design an object detection model: *single shot multibox detection* (*SSD*)
- this model is simple, fast, and widely used
- although this is just one of vast amounts of object detection models, some of the design principles and implementation details in this section are also applicable to other models

6.6 Single shot multibox detection

- Figure 4 provides an overview of the design of single-shot multibox detection
- this model mainly consists of a base network, followed by several multiscale feature map blocks
- the base network is for extracting features from the input image, so it can use a deep CNN
- for example, the original single-shot multibox detection paper adopts a VGG network truncated before the classification layer, while ResNet has also been commonly used
- through our design, we can make the base network output larger feature maps, so as to generate more anchor boxes for detecting smaller objects
- subsequently, each multiscale feature map block reduces (e.g., by half) the height and width of the feature maps from the previous block, and enables each unit of the feature maps to increase its receptive field on the input image

6.6 Single shot multibox detection

- recall the design of multiscale object detection through layer-wise representations of images by deep neural networks, in Section 6.5
- since multiscale feature maps closer to the top of Figure 4 are smaller, but have larger receptive fields, they are suitable for detecting fewer, but larger objects
- in a nutshell, via its base network and several multiscale feature map blocks, single-shot multibox detection generates a varying number of anchor boxes with different sizes, and detects varying-size objects by predicting classes and offsets of these anchor boxes (thus the bounding boxes); thus, this is a multiscale object detection model

6.6 Single shot multibox detection

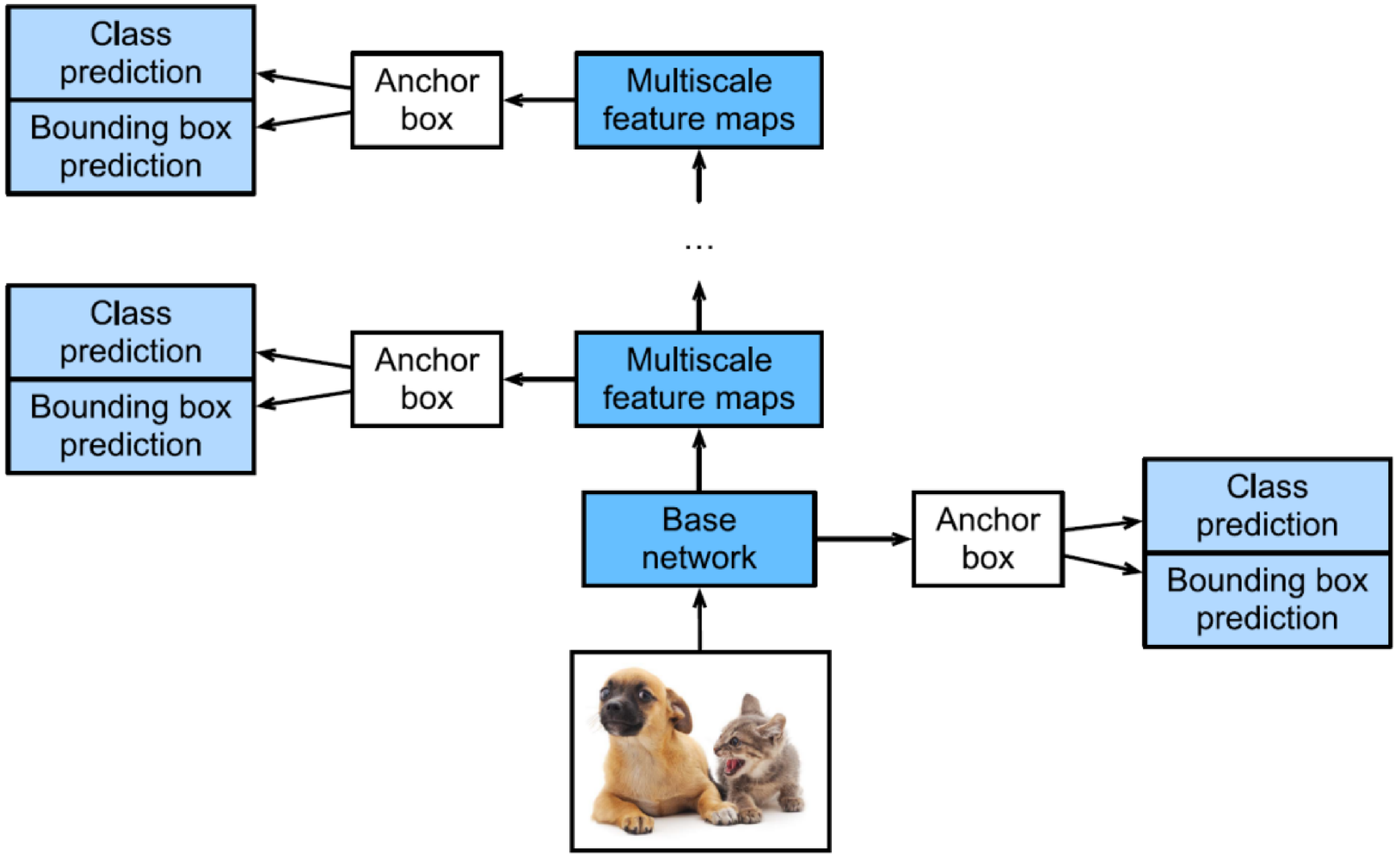


Figure 4: As a multiscale object detection model, single-shot multibox detection mainly consists of a base network followed by several multiscale feature map blocks.

6.6 Single shot multibox detection

- in the following, we will describe the details of different blocks in Figure 4
- to begin with, we discuss how to implement the class and bounding box prediction
- let the number of object classes be q; then, anchor boxes have $q + 1$ classes, where class 0 is background
- at some scale, suppose that the height and width of feature maps are h and w, respectively
- when a anchor boxes are generated with each spatial position of these feature maps as their center, a total of hwa anchor boxes need to be classified
- this often makes classification with fully-connected layers infeasible, due to likely heavy parameterization costs
- single-shot multibox detection uses channels of convolutional layers to predict classes, to reduce model complexity

6.6 Single shot multibox detection

- specifically, the class prediction layer uses a convolutional layer, without altering the width or height of the feature maps
- in this way, there can be a one-to-one correspondence between outputs and inputs at the same spatial dimensions (width and height) of feature maps
- more concretely, channels of the output feature maps at any spatial position (x, y) represent class predictions for all the anchor boxes centered on (x, y) of the input feature maps
- to produce valid predictions, there must be $a(q + 1)$ output channels, where, for the same spatial position, the output channel with index $i(q + 1) + j$ represents the prediction of the class j $(0 \leq j \leq q)$ for the anchor box i $(0 \leq i < a)$

6.6 Single shot multibox detection

- we can define such a class prediction layer, to use a 3×3 convolutional layer with a padding of 1
- the width and height of the input and output of this convolutional layer remain unchanged
- the design of the bounding box prediction layer is similar to that of the class prediction layer
- the only difference lies in the number of outputs for each anchor box: here, we need to predict four offsets, rather than $q + 1$ classes
- as we mentioned, single-shot multibox detection uses multiscale feature maps to generate anchor boxes and predict their classes and offsets
- at different scales, the shapes of feature maps or the numbers of anchor boxes centered on the same unit may vary
- therefore, shapes of the prediction outputs at different scales may vary

6.6 Single shot multibox detection

- in order to detect objects at multiple scales, we define a downsampling block that halves the height and width of input feature maps
- in fact, this block applies the design of VGG blocks in Section 3.8
- more concretely, each downsampling block consists of two 3×3 convolutional layers with padding of 1, followed by a 2×2 maximum pooling layer with stride of 2
- as we know, 3×3 convolutional layers with padding of 1 do not change the shape of feature maps
- however, the subsequent 2×2 maximum pooling reduces the height and width of input feature maps by half
- for both input and output feature maps of this downsampling block, because $1 \times 2 + (3 - 1) + (3 - 1) = 6$, each unit in the output has a 6×6 receptive field on the input
- therefore, the downsampling block enlarges the receptive field of each unit in its output feature maps

6.6 Single shot multibox detection

- the base network block is then used to extract features from input images
- for simplicity, we construct a small base network consisting of three downsampling blocks that double the number of channels at each block
- given a 256×256 input image, this base network block outputs 32×32 feature maps ($256/2^3 = 32$)
- the complete single shot multibox detection model consists of five blocks
- the feature maps produced by each block are used for both (i) generating anchor boxes and (ii) predicting classes and offsets of these anchor boxes
- among these five blocks, the first one is the base network block, the second to the fourth are downsampling blocks, and the last block uses global maximum pooling to reduce both the height and width to 1
- technically, the second to the fifth blocks are all those multiscale feature map blocks in Figure 4

6.6 Single shot multibox detection

- different from the case of image classification tasks, the outputs of each block include (i) CNN feature maps **Y**, (ii) anchor boxes generated using **Y** at the current scale, and (iii) classes and offsets predicted (based on **Y**) for these anchor boxes
- recall that, in Figure 4, a multiscale feature map block that is closer to the top is for detecting larger objects; thus, it needs to generate larger anchor boxes
- the interval between 0.2 and 1.05 has to be split evenly into five sections, to determine the smaller scale values at the five blocks: 0.2, 0.37, 0.54, 0.71, and 0.88
- then, their larger scale values are given by $\sqrt{0.2 \times 0.37} = 0.272$, $\sqrt{0.37 \times 0.54} = 0.447$, and so on

6.6 Single shot multibox detection

- for example, let's say we apply the model on a mini-batch of 256×256 images
- as explained earlier in this section, the first block outputs 32×32 feature maps
- recall that the second to fourth downsampling blocks halve the height and width, and the fifth block uses global pooling
- since 4 anchor boxes are generated for each unit along spatial dimensions of feature maps, at all the five scales, a total of $(32^2 + 16^2 + 8^2 + 4^2 + 1) \times 4 = 5444$ anchor boxes are generated for each image
- now, we will explain how to train the single shot multibox detection model for object detection

6.6 Single shot multibox detection

- object detection has two types of losses
- the first loss concerns classes of anchor boxes: its computation can simply reuse the cross-entropy loss function that we used for image classification
- the second loss concerns offsets of positive (non-background) anchor boxes: this is a regression problem
- however, for this regression problem, here, we do not use the squared loss
- instead, we use the ℓ_1 norm loss, the absolute value of the difference between the prediction and the ground-truth
- as discussed in Section 6.4, we filter out negative anchor boxes and illegal (padded) anchor boxes in the loss calculation
- in the end, we sum up the anchor box class loss and the anchor box offset loss, to obtain the loss function for the model

6.6 Single shot multibox detection

- we can use accuracy to evaluate the classification results
- due to the used ℓ_1 norm loss for the offsets, we can use the *mean absolute error* to evaluate the predicted bounding boxes
- these prediction results are obtained from the generated anchor boxes and the predicted offsets for them
- when training the model, we need to generate multiscale anchor boxes and predict their classes and offsets in the forward propagation
- then, we label the classes and offsets of such generated anchor boxes based on the label information

6.6 Single shot multibox detection

- finally, we calculate the loss function using the predicted and labeled values of the classes and offsets
- during prediction, the goal is to detect all the objects of interest on the image
- the predicted bounding boxes are then obtained from the anchor boxes and their predicted offsets
- then, non-maximum suppression is used to remove similar predicted bounding boxes
- finally, we display all the predicted bounding boxes with confidence 0.9 or above, as the output